

Blood Bank Inventory & Donor Matcher —

Abstract

The Blood Bank Inventory & Donor Matcher is a software application designed to simplify and automate the management of blood stocks, donor records, borrowing/issuing activities, and compatibility matching in a hospital or community blood bank. Traditional blood inventory management methods often rely on manual paper logbooks, which can be time-consuming, error-prone, difficult to maintain, and dangerous during emergency situations. This system provides a centralized platform for efficiently managing life-saving blood resources and donor information.

The system allows administrators and healthcare staff to add, update, search, and remove donor records, maintaining details such as donor ID, donor name, blood group, age, phone number, and eligibility status. It also manages real-time blood stock levels across all major blood types (A+, A-, B+, B-, AB+, AB-, O+, O-), tracking available units, batch IDs, and expiration dates. Users can easily search for compatible blood units and view available stock. The system can execute donor-recipient compatibility logic, track blood issuances, and help staff identify low-stock levels or expiring blood bags, improving the efficiency and safety of the blood distribution process.

By automating routine operations, the proposed system reduces paperwork, minimizes data-entry errors, saves time, and improves accessibility of vital healthcare information. It also provides an organized and reliable method for maintaining records and generating useful inventory reports. The Blood Bank Inventory & Donor Matcher can be implemented as a web or desktop application (or console system) and can be further enhanced with features such as SMS donor dispatch alerts, online blood reservations, automated eligibility checking, and role-based access. Overall, the system aims to provide a simple, efficient, accurate, and user-friendly solution for managing blood bank activities.

Simple Roadmap

1. Choose Technology

Use: Java

2. Main Menu

When the program starts, display:

```
===== BLOOD BANK INVENTORY & DONOR MATCHER =====  
1. Add Donor  
2. View Donors  
3. Search Donor by Blood Group  
4. Add / Update Blood Stock  
5. View Blood Stock
```

6. Request & Match Blood (Issue Unit)
7. Record Blood Donation
8. Exit

3. Create Basic Classes

Create these classes:

- Donor
- BloodInventory
- BloodBank
- Main

Donor

- Donor ID
- Donor Name
- Blood Group
- Age
- Phone Number

BloodInventory

- Blood Group
- Units Available
- Expiration Date
- Batch ID

BloodBank

- Add donor
- Search donor
- Update stock
- Match blood
- Issue unit

Main

- Display menu
- Get user choice
- Call required methods

4. Develop Step-by-Step

Step 1: Add Donor

Enter Donor ID:

Enter Donor Name:

Enter Age:

Enter Blood Group:

Enter Phone Number:

Donor added successfully!

Step 2: View Donors

Display all registered donors with their blood group and contact details.

Step 3: Search Donor

User enters Blood Group (e.g. O-) → display matching donor details.

Step 4: Add / Update Blood Stock

Enter Blood Group:

Enter Units to Add:

Enter Expiration Date:

Blood stock updated successfully!

Step 5: View Blood Stock

Display available units and expiration dates for each blood group.

Step 6: Request & Match Blood

Enter Recipient Blood Group:

Enter Units Required:

Blood matched and issued successfully!

Change stock status from:

Available → Issued

Step 7: Record Blood Donation

Enter Donor ID:

Enter Units Donated:

Donation recorded successfully!

Stock incremented!

5. Testing

Test each option:

- Add 5 donors with varied blood groups
- Add initial blood stock for all blood types
- Search donors by blood group (e.g., O- or A+)
- Issue a blood unit for a compatible request
- Try issuing an out-of-stock or incompatible blood unit
- Record a new donation and verify stock update
- Try registering an ineligible donor (< 56 days since last donation)

6. Final Project Structure

BloodBankManagement/

Donor.java

BloodInventory.java

BloodBank.java

Main.java